

Quantum Entanglement & Quantum Computing

Units 1 & 2 Question Bank

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UE23EC343AB3: Quantum Entanglement & Quantum Computing

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UNIT 1: BASICS OF QUANTUM TECH

1-Mark Multiple Choice Questions (20 Questions)

Question 1.1.1 In the double-slit experiment performed with bullets, the pattern observed on the screen is:

- A An interference pattern with fringes
- B The simple sum of the two single-slit patterns
- C A single central spot only
- D A uniform illumination

Answer: B

Solution (1 mark): Classical particles carry no phase. Probabilities add directly, $P_{12} = P_1 + P_2$, giving no fringes.

Question 1.1.2 For the double-slit experiment with waves, the correct combination rule is:

- A $I_{12} = I_1 + I_2$
- B $I_{12} = |h_1 + h_2|^2$
- C $I_{12} = |h_1|^2 + |h_2|^2$
- D $I_{12} = h_1 h_2$

Answer: B

Solution (1 mark): Amplitudes add, then the intensity is the modulus square. The cross term $2\text{Re}(h_1 h_2^*)$ is the interference term.

Question 1.1.3 Electrons fired one at a time through a double slit produce:

- A Two bands, like bullets
- B An interference pattern built up dot by dot
- C A continuous smear with no structure
- D No detection events

Answer: B

Solution (1 mark): Each electron arrives as a localized dot, but the accumulated distribution follows $|\psi_1 + \psi_2|^2$. Each electron interferes with itself.

Question 1.1.4 If a which-path detector is placed at the slits in the electron experiment:

- A The fringes become sharper
- B The interference pattern disappears
- C The electrons stop arriving
- D The pattern shifts sideways

Answer: B

Solution (1 mark): Path information destroys coherence between the two amplitudes. The state becomes a mixture and probabilities add classically.

Question 1.1.5 In Dirac notation, $\langle\psi|$ denotes:

- A A vector in the Hilbert space $\mathcal{H}$
- B A linear functional in the dual space $\mathcal{H}^*$
- C An observable
- D A probability

Answer: B

Solution (1 mark): The bra is the Hermitian conjugate of the ket and acts on kets to give a complex number, $\langle\psi|\phi\rangle \in \mathbb{C}$.

Question 1.1.6 The normalization condition for a quantum state is:

- A $\langle\psi|\psi\rangle = 0$
- B $\langle\psi|\psi\rangle = 1$
- C $|\psi\rangle = 1$
- D $\text{Tr}(|\psi\rangle) = 1$

Answer: B

Solution (1 mark): Total probability over all outcomes must equal one, so $\sum_i |c_i|^2 = \langle \psi | \psi \rangle = 1$.

Question 1.1.7 A Hilbert space is:

- A A real vector space with a cross product
- B A complete complex inner-product space
- C Any set of matrices
- D A space of classical probability distributions

Answer: B

Solution (1 mark): Quantum states live in a complex vector space equipped with an inner product and complete with respect to the induced norm.

Question 1.1.8 Observables in quantum mechanics are represented by:

- A Unitary operators
- B Hermitian operators
- C Arbitrary complex matrices
- D Projectors only

Answer: B

Solution (1 mark): $A = A^\dagger$ guarantees real eigenvalues (measurable) and an orthonormal eigenbasis (spectral theorem).

Question 1.1.9 Time evolution of a closed quantum system is described by an operator that is:

- A Hermitian
- B Unitary
- C Singular
- D Anti-symmetric and real

Answer: B

Solution (1 mark): $U = e^{-iHt/\hbar}$ with $U^\dagger U = I$. Unitarity preserves the norm, hence total probability.

Question 1.1.10 The linearity of quantum mechanics implies that if $|\psi_1\rangle$ and $|\psi_2\rangle$ are valid states, then:

- A Only their sum is valid
- B $\alpha|\psi_1\rangle + \beta|\psi_2\rangle$ is valid after normalization
- C Their product is valid
- D Neither is valid together

Answer: B

Solution (1 mark): The superposition principle: any normalized complex linear combination of physical states is itself a physical state.

Question 1.1.11 The completeness relation for a discrete orthonormal basis $\{|i\rangle\}$ is:

- A $\sum_i |i\rangle\langle i| = I$
- B $\sum_i \langle i|i\rangle = 0$
- C $\sum_i |i\rangle = I$
- D $\langle i|j\rangle = 1$ for all i, j

Answer: A

Solution (1 mark): The resolution of identity. Inserting it lets any state be expanded as $|\psi\rangle = \sum_i \langle i|\psi\rangle |i\rangle$.

Question 1.1.12 For a continuous basis, the orthogonality condition reads:

- A $\langle x|x'\rangle = \delta_{xx'}$
- B $\langle x|x'\rangle = \delta(x - x')$
- C $\langle x|x'\rangle = 1$
- D $\langle x|x'\rangle = 0$ always

Answer: B

Solution (1 mark): Continuous bases use the Dirac delta, with completeness $\int dx |x\rangle\langle x| = I$. Such states are not normalizable in the usual sense.

Question 1.1.13 The quantity $|\psi(x)|^2$ represents:

- A Probability of finding the particle at exactly x
- B Probability density per unit length
- C The energy at x
- D The momentum at x

Answer: B

Solution (1 mark): For continuous variables the probability of finding the particle in $[x, x + dx]$ is $|\psi(x)|^2 dx$.

Question 1.1.14 The expectation value of an observable A in state $|\psi\rangle$ is:

- A $\langle\psi|A|\psi\rangle$
- B $|\langle\psi|A|\psi\rangle|^2$
- C $A|\psi\rangle$
- D $\text{Tr}(A)$

Answer: A

Solution (1 mark): $\langle A \rangle = \langle\psi|A|\psi\rangle = \sum_a a p_a$, the mean of the measurement outcomes weighted by the Born probabilities.

Question 1.1.15 The matrix representation of $|0\rangle$ in the computational basis is:

- A $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
- B $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- C $\begin{pmatrix} 1 & 0 \end{pmatrix}$
- D $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Answer: B

Solution (1 mark): By convention $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$. A row vector would be the bra $\langle 0|$.

Question 1.1.16 A general pure qubit state on the Bloch sphere is written as:

- A $\cos \theta |0\rangle + \sin \theta |1\rangle$
- B $\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$
- C $e^{i\theta} |0\rangle + e^{i\phi} |1\rangle$
- D $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Answer: B

Solution (1 mark): Two real parameters (θ, ϕ) remain after fixing normalization and discarding the global phase. The half-angle reflects the $2 \rightarrow 1$ map from spinors to the sphere.

Question 1.1.17 Antipodal points on the Bloch sphere correspond to states that are:

- A Identical
- B Orthogonal
- C Maximally mixed
- D Non-normalizable

Answer: B

Solution (1 mark): $\theta \rightarrow \pi - \theta$, $\phi \rightarrow \phi + \pi$ maps a state to its orthogonal complement. Example: $|0\rangle$ (north pole) and $|1\rangle$ (south pole).

Question 1.1.18 A global phase $e^{i\gamma}|\psi\rangle$:

- A Changes all measurement probabilities
- B Has no observable consequence
- C Changes only relative phases
- D Destroys normalization

Answer: B

Solution (1 mark): $|\langle a|e^{i\gamma}\psi\rangle|^2 = |\langle a|\psi\rangle|^2$. Only relative phases within a superposition are physical.

Question 1.1.19 The canonical commutation relation is:

- A $[\hat{x}, \hat{p}] = 0$
- B $[\hat{x}, \hat{p}] = i\hbar$
- C $[\hat{x}, \hat{p}] = \hbar$
- D $[\hat{x}, \hat{p}] = -1$

Answer: B

Solution (1 mark): Non-commutation leads to the uncertainty relation $\Delta x \Delta p \geq \hbar/2$. Position and momentum have no common eigenbasis.

Question 1.1.20 In Qiskit, the statevector of a freshly created 2-qubit QuantumCircuit before any gate is:

- A $|00\rangle$
- B $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$
- C Random
- D $|11\rangle$

Answer: A

Solution (1 mark): All qubits initialize to $|0\rangle$, so the register starts in $|0\rangle^{\otimes n}$. Superposition requires explicit Hadamard gates.

True/False Style Questions (20 Questions)

Question 1.1.21 Statement: "A quantum state vector must always be normalized to have unit length." This statement is:

- A True
- B False
- C True only for mixed states
- D True only in the momentum basis

Answer: A

Solution (1 mark): True. $\langle\psi|\psi\rangle = 1$ so that total probability across all outcomes sums to one.

Question 1.1.22 Statement: “Two orthogonal quantum states can always be perfectly distinguished by a single measurement.” This statement is:

- A True
- B False
- C Only true for mixed states
- D Only true for entangled states

Answer: A

Solution (1 mark): True. Orthogonal states have projectors with zero overlap, so a suitable projective measurement identifies them with certainty.

Question 1.1.23 Statement: “The global phase of a quantum state is experimentally measurable.” This statement is:

- A True
- B False
- C True only for qubits
- D True only after measurement

Answer: B

Solution (1 mark): False. $|\langle a|e^{i\gamma}\psi\rangle|^2 = |\langle a|\psi\rangle|^2$; global phase drops out of every measurement probability.

Question 1.1.24 Statement: “Every Hermitian operator has real eigenvalues.” This statement is:

- A True
- B False
- C True only for unitary operators
- D True only for 2×2 matrices

Answer: A

Solution (1 mark): True. This is guaranteed by the spectral theorem, which is why observables are represented by Hermitian operators.

Question 1.1.25 Statement: “A qubit in state $|0\rangle$ and one in state $|1\rangle$ can be perfectly cloned by a universal machine.” This statement is:

- A True
- B False
- C True only for entangled qubits
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Known, distinguishable basis states can be copied deterministically. The no-cloning theorem forbids cloning arbitrary *unknown* states.

Question 1.1.26 Statement: “Time evolution of a closed quantum system can change the norm of the state vector.” This statement is:

- A True
- B False
- C True only for degenerate Hamiltonians
- D True only at $t = 0$

Answer: B

Solution (1 mark): False. Evolution is unitary, $U^\dagger U = I$, which strictly preserves the norm at all times.

Question 1.1.27 Statement: “In the double-slit experiment, firing electrons one at a time destroys the interference pattern.” This statement is:

- A True
- B False
- C True only for photons
- D True only above room temperature

Answer: B

Solution (1 mark): False. The pattern still builds up dot by dot; each electron interferes with itself via the two path amplitudes.

Question 1.1.28 Statement: “Placing a which-path detector at the slits destroys the interference fringes.” This statement is:

- A True
- B False
- C Only if the detector is switched off
- D Only for massive particles

Answer: A

Solution (1 mark): True. Which-path information decoheres the two amplitudes, so probabilities add classically and fringes vanish.

Question 1.1.29 Statement: “A ket $|\psi\rangle$ and its bra $\langle\psi|$ live in the same vector space.” This statement is:

- A True
- B False
- C True only for real coefficients
- D True only for eigenstates

Answer: B

Solution (1 mark): False. The bra is an element of the dual space $\mathcal{H}^*$, not $\mathcal{H}$ itself.

Question 1.1.30 Statement: “The Bloch sphere can represent the state of any single qubit, pure or mixed.” This statement is:

- A True
- B False
- C True only for pure states
- D True only for entangled states

Answer: A

Solution (1 mark): True. Pure states sit on the surface ($|\vec{r}| = 1$), mixed states sit strictly inside ($|\vec{r}| < 1$).

Question 1.1.31 Statement: “Position and momentum operators commute in quantum mechanics.” This statement is:

- A True
- B False
- C True only in 1D
- D True only for free particles

Answer: B

Solution (1 mark): False. $[\hat{x}, \hat{p}] = i\hbar \neq 0$, which is the origin of the position-momentum uncertainty relation.

Question 1.1.32 Statement: “ $|\langle 0|1\rangle| = 0$ in the computational basis.” This statement is:

- A True
- B False
- C True only after measurement
- D Cannot be determined

Answer: A

Solution (1 mark): True. $|0\rangle$ and $|1\rangle$ are orthonormal computational basis states, so their overlap is zero.

Question 1.1.33 Statement: “A freshly initialized n -qubit Qiskit register starts in an equal superposition of all basis states.” This statement is:

- A True
- B False
- C True only for $n = 1$
- D True only after measurement

Answer: B

Solution (1 mark): False. It starts in $|0\rangle^{\otimes n}$; superposition requires explicit Hadamard gates.

Question 1.1.34 Statement: “Linearity implies that any complex linear combination of two valid states, once normalized, is itself a valid state.” This statement is:

- A True
- B False
- C True only for orthogonal states
- D True only for real coefficients

Answer: A

Solution (1 mark): True. This is the superposition principle, one of the foundational postulates of quantum mechanics.

Question 1.1.35 Statement: “ $|\psi(x)|^2$ gives the exact position of a particle.” This statement is:

- A True
- B False
- C True only for eigenstates of $\hat{x}$
- D True only in the classical limit

Answer: B

Solution (1 mark): False. $|\psi(x)|^2$ is a probability *density*; the probability of being in $[x, x + dx]$ is $|\psi(x)|^2 dx$.

Question 1.1.36 Statement: “A continuous basis such as $\{|x\rangle\}$ consists of normalizable states.” This statement is:

- A True
- B False
- C True only for bound states
- D True only for momentum eigenstates

Answer: B

Solution (1 mark): False. Position eigenstates satisfy $\langle x|x'\rangle = \delta(x - x')$, which is not normalizable in the usual sense.

Question 1.1.37 Statement: “Antipodal points on the Bloch sphere always represent orthogonal states.” This statement is:

- A True
- B False
- C Only for real-amplitude states
- D Only for mixed states

Answer: A

Solution (1 mark): True. $\theta \rightarrow \pi - \theta$, $\phi \rightarrow \phi + \pi$ always maps a state to its orthogonal complement, e.g. $|0\rangle$ and $|1\rangle$.

Question 1.1.38 Statement: “The expectation value $\langle A \rangle = \langle \psi|A|\psi \rangle$ can be a complex number for a Hermitian A .” This statement is:

- A True
- B False
- C True only for mixed states
- D True only for degenerate spectra

Answer: B

Solution (1 mark): False. Hermiticity guarantees $\langle A \rangle$ is always real, since it is a weighted average of real eigenvalues.

Question 1.1.39 Statement: “The resolution of identity $\sum_i |i\rangle\langle i| = I$ holds for any orthonormal basis.” This statement is:

- A True
- B False
- C True only for the computational basis
- D True only for 2-level systems

Answer: A

Solution (1 mark): True. Completeness is a property of any orthonormal basis of the Hilbert space, not a special one.

Question 1.1.40 Statement: “A Hilbert space used in quantum mechanics must be finite-dimensional.” This statement is:

- A True
- B False
- C True only for qubit systems
- D True only for composite systems

Answer: B

Solution (1 mark): False. Qubits use finite-dimensional spaces, but position/momentum systems require infinite-dimensional Hilbert spaces.

Additional Easy Recall Questions (63 Questions)

Question 1.1.41 Statement: “The time-dependent Schrodinger equation is $i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This is the fundamental equation governing how a quantum state evolves in time.

Question 1.1.42 Statement: “The wavefunction $\psi(x)$ of a particle must always be a real number.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: B

Solution (1 mark): False. Wavefunctions are generally complex-valued; only $|\psi(x)|^2$ (a probability density) must be real and non-negative.

Question 1.1.43 Statement: “The momentum operator in the position representation is $\hat{p} = -i\hbar \frac{d}{dx}$.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This is the standard representation used to convert the classical momentum into a quantum operator.

Question 1.1.44 Statement: “A particle in an infinite square well can have zero total energy.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. The ground state energy $E_1 = \pi^2 \hbar^2 / 2mL^2$ is strictly positive; zero energy would violate the uncertainty principle.

Question 1.1.45 Statement: “Every eigenstate of the Pauli Z operator is also an eigenstate of the Pauli X operator.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: B

Solution (1 mark): False. X and Z do not commute, so they share no common eigenbasis (except in trivial cases).

Question 1.1.46 Statement: “The eigenvalues of the Pauli Z matrix are $+1$ and -1 .” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. $Z|0\rangle = +|0\rangle$ and $Z|1\rangle = -|1\rangle$, so $|0\rangle, |1\rangle$ are its eigenstates with these eigenvalues.

Question 1.1.47 Statement: “ $|0\rangle$ and $|1\rangle$ are eigenstates of the Pauli Z operator.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. By convention, the computational basis states are defined as the eigenstates of Z .

Question 1.1.48 Statement: “ $|+\rangle$ and $|-\rangle$ are eigenstates of the Pauli X operator.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. $X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$, so they diagonalize X .

Question 1.1.49 Statement: “The Pauli matrices X, Y, Z all square to the identity matrix.” This statement is:

- A True

- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. $X^2 = Y^2 = Z^2 = I$; this is a defining algebraic property of the Pauli operators.

Question 1.1.50 Statement: “The trace of any single Pauli matrix (X , Y , or Z) is zero.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Each Pauli matrix has eigenvalues $+1$ and -1 , which sum to zero.

Question 1.1.51 Statement: “The north pole of the Bloch sphere represents the state $|0\rangle$.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. By convention $\theta = 0$ (north pole) corresponds to $|0\rangle$.

Question 1.1.52 Statement: “The south pole of the Bloch sphere represents the state $|1\rangle$.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. By convention $\theta = \pi$ (south pole) corresponds to $|1\rangle$.

Question 1.1.53 Statement: “Points on the equator of the Bloch sphere represent equal-weight superpositions of $|0\rangle$ and $|1\rangle$.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. At $\theta = \pi/2$, $\cos(\theta/2) = \sin(\theta/2)$, giving equal amplitudes for both basis states.

Question 1.1.54 Statement: “A single qubit requires a real (not complex) vector space to be fully described.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: B

Solution (1 mark): False. A complex Hilbert space is required; relative phases between amplitudes are physically meaningful and need complex numbers.

Question 1.1.55 Statement: “Two qubits together live in a 4-dimensional Hilbert space.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. The joint space is the tensor product of two 2-dimensional spaces, giving dimension $2 \times 2 = 4$.

Question 1.1.56 Statement: “An n -qubit system lives in a Hilbert space of dimension 2^n .” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Each additional qubit doubles the dimension via the tensor product structure.

Question 1.1.57 Statement: “Quantum operators act linearly on state vectors.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Linearity is a foundational requirement so that superpositions transform consistently.

Question 1.1.58 Statement: “The inner product $\langle\psi|\phi\rangle$ is always a real number.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. In general it is a complex number; it is real only in special cases such as $\langle\psi|\psi\rangle$.

Question 1.1.59 Statement: “The adjoint of a product of operators reverses their order: $(AB)^\dagger = B^\dagger A^\dagger$.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This is a standard property of the Hermitian conjugate for operator products.

Question 1.1.60 Statement: “If $A|a\rangle = a|a\rangle$, then $|a\rangle$ is called an eigenstate of A with eigenvalue a .” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is simply the definition of an eigenvalue equation.

Question 1.1.61 Statement: “A projector P satisfies $P^2 = P$.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Idempotency is the defining algebraic property of a projection operator.

Question 1.1.62 Statement: “The outer product $|a\rangle\langle a|$ for a normalized $|a\rangle$ is a valid projector.” This statement is:

- A True

- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. $(|a\rangle\langle a|)^2 = |a\rangle\langle a|a\rangle\langle a| = |a\rangle\langle a|$ when $\langle a|a\rangle = 1$.

Question 1.1.63 Statement: “A single qubit has exactly 2 computational basis states.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. These are $|0\rangle$ and $|1\rangle$.

Question 1.1.64 Statement: “Two qubits have exactly 4 computational basis states.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. These are $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Question 1.1.65 Statement: “The relative phase between terms in a superposition has no physical effect.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. Relative phase is physically meaningful and affects interference; only the overall global phase is unobservable.

Question 1.1.66 Statement: “Measuring a qubit in the computational basis means measuring in the eigenbasis of Z .” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. The computational basis $\{|0\rangle, |1\rangle\}$ is defined as the Z -eigenbasis.

Question 1.1.67 Statement: “Photons exhibit both wave-like and particle-like behaviour.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This wave-particle duality is one of the foundational observations of quantum mechanics.

Question 1.1.68 Statement: “The de Broglie wavelength of a particle is given by $\lambda = h/p$.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This relates a particle’s momentum p to an associated wavelength λ , with h the Planck constant.

Question 1.1.69 Statement: “The position operator $\hat{x}$ acts on a wavefunction by simple multiplication by x .” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. In the position representation, $\hat{x} \psi(x) = x \psi(x)$.

Question 1.1.70 Statement: “The commutator of an operator with itself, $[\hat{A}, \hat{A}]$, is always zero.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Any operator trivially commutes with itself.

Question 1.1.71 Statement: “If a system is prepared in an eigenstate of an observable, measuring that observable again immediately afterward gives the same eigenvalue.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. The first measurement collapses the state into the eigenstate, so a repeated measurement is deterministic.

Question 1.1.72 Statement: “Quantum mechanics gives fully deterministic predictions for the outcome of every individual measurement.” This statement is:

- A True

- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. Quantum mechanics is fundamentally probabilistic; it predicts probabilities of outcomes, not individual results (except for eigenstates of the measured observable).

Question 1.1.73 Statement: “The Schrodinger equation is a linear differential equation in the state $|\psi\rangle$.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Linearity of the Schrodinger equation is what makes the superposition principle consistent under time evolution.

Question 1.1.74 Statement: “A quantum state remains in superposition until a measurement is performed.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is the standard measurement postulate: superpositions persist under unitary evolution and collapse only upon measurement.

Question 1.1.75 Statement: “The Stern-Gerlach experiment demonstrated that electron spin is quantized.” This statement is:

- A True
- B False

C True only above room temperature

D False only at zero temperature

Answer: A

Solution (1 mark): True. It showed silver atoms deflect into two discrete spots rather than a continuous spread, revealing quantized spin angular momentum.

Question 1.1.76 Statement: “A probability amplitude and a probability are the same thing.” This statement is:

A True

B False

C Cannot be determined

D True only in the Heisenberg picture

Answer: B

Solution (1 mark): False. A probability amplitude is a complex number; the probability is its squared magnitude.

Question 1.1.77 Statement: “An orthonormal basis consists of vectors that are mutually orthogonal and individually normalized.” This statement is:

A True

B False

C True only in the classical limit

D False except for entangled states

Answer: A

Solution (1 mark): True. This is the definition of an orthonormal set: $\langle i | j \rangle = \delta_{ij}$.

Question 1.1.78 Statement: “Spin-1/2 particles, like electrons, can be described using the same two-level qubit formalism.” This statement is:

A True

B False

C True only for qubits

D False only after measurement

Answer: A

Solution (1 mark): True. A spin-1/2 particle's spin-up and spin-down states form a natural two-level (qubit-like) system.

Question 1.1.79 Statement: "Once a qubit is measured in a basis, the superposition in that basis is destroyed." This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Measurement collapses the state to one of the basis eigenstates, eliminating the original superposition.

Question 1.1.80 Statement: "An arbitrary unknown quantum state can be measured without disturbing it in any way." This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: B

Solution (1 mark): False. Measurement generally disturbs an unknown state (unless it happens to be an eigenstate of the measured observable).

Question 1.1.81 Statement: "The matrix representation of an operator depends on the choice of basis." This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. The same abstract operator has different matrix forms in different bases; only the underlying operator is basis-independent.

Question 1.1.82 Statement: “The trace of the 2×2 identity matrix is 2.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. $\text{Tr}(I_2) = 1 + 1 = 2$.

Question 1.1.83 Statement: “A global phase factor $e^{i\gamma}$ changes the sign of the probability of every measurement outcome.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: B

Solution (1 mark): False. A global phase leaves all measurement probabilities completely unchanged, since $|e^{i\gamma}| = 1$.

Question 1.1.84 Statement: “The identity operator has no effect when applied to any quantum state.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. By definition, $I|\psi\rangle = |\psi\rangle$ for every state.

Question 1.1.85 Statement: “In Dirac notation, $|\psi\rangle\langle\phi|$ represents an operator, not a number.” This statement is:

- A True

- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This outer product construction yields a linear operator, in contrast to the inner product $\langle\phi|\psi\rangle$ which is a scalar.

Question 1.1.86 Statement: “A qubit can only be in the state $|0\rangle$ or $|1\rangle$, never both at once.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. A qubit can exist in a superposition $\alpha|0\rangle + \beta|1\rangle$ of both basis states simultaneously.

Question 1.1.87 Statement: “The uncertainty principle states that position and momentum can be measured simultaneously with arbitrary precision.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: B

Solution (1 mark): False. The uncertainty principle states the opposite: $\Delta x \Delta p \geq \hbar/2$, a fundamental limit on joint precision.

Question 1.1.88 Statement: “A Hermitian operator is always equal to its own adjoint, $A = A^\dagger$.” This statement is:

- A True
- B False

- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is the defining property of a Hermitian (self-adjoint) operator.

Question 1.1.89 Statement: “Every unitary operator is also Hermitian.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: B

Solution (1 mark): False. Unitary means $U^\dagger U = I$; this does not generally imply $U = U^\dagger$ (though some special unitaries, like the Pauli matrices, are both).

Question 1.1.90 Statement: “The Pauli Y matrix has purely imaginary off-diagonal entries.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, so its off-diagonal entries are $\pm i$.

Question 1.1.91 Statement: “A quantum state vector can have a negative norm.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: B

Solution (1 mark): False. The norm $\langle\psi|\psi\rangle$ is always non-negative, since it is a sum of squared magnitudes.

Question 1.1.92 Statement: “In the double-slit experiment, closing one slit removes the possibility of interference.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. With only one slit open there is only one path amplitude, so there is nothing for it to interfere with.

Question 1.1.93 Statement: “The probability of measuring a particular outcome can be negative in standard quantum mechanics.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. The Born rule gives probabilities as $|amplitude|^2$, which is always non-negative.

Question 1.1.94 Statement: “A ket $|\psi\rangle$ can be expanded in any complete orthonormal basis of the Hilbert space.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Completeness of the basis guarantees that any state vector can be written as a linear combination of the basis states.

Question 1.1.95 Statement: “The energy eigenstates of a time-independent Hamiltonian evolve only by acquiring a phase factor over time.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. For $H|E_n\rangle = E_n|E_n\rangle$, the state evolves as $|E_n\rangle \rightarrow e^{-iE_nt/\hbar}|E_n\rangle$.

Question 1.1.96 Statement: “Two quantum states with the same $|\langle\psi|\phi\rangle|$ have the same physical distinguishability.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Distinguishability depends only on the magnitude of the overlap, which determines the measurement statistics.

Question 1.1.97 Statement: “A qubit’s state can be fully specified using only classical bits and no complex numbers.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: B

Solution (1 mark): False. In general a qubit’s amplitudes are complex numbers; classical bits cannot capture the phase information.

Question 1.1.98 Statement: “The identity relation $\sum_i |i\rangle\langle i| = I$ only holds for the computational basis.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: B

Solution (1 mark): False. It holds for any complete orthonormal basis, not just the computational one.

Question 1.1.99 Statement: “Any two orthogonal quantum states have zero inner product.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This is the definition of orthogonality: $\langle\psi|\phi\rangle = 0$.

Question 1.1.100 Statement: “The number of real parameters needed to specify a normalized single-qubit state (up to global phase) is 2.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. These are the Bloch angles θ and ϕ .

Question 1.1.101 Statement: “Quantum superposition is a phenomenon unique to subatomic particles and cannot be described mathematically for any two-level system.” This statement is:

- A True

- B False
- C True only in Qiskit
- D False only on real hardware

Answer: B

Solution (1 mark): False. Superposition is a general mathematical feature of any two-level (or higher) quantum system, including engineered qubits.

Question 1.1.102 Statement: “The energy operator (Hamiltonian) for a free particle contains only a kinetic energy term.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. With no potential, $H = \hat{p}^2/2m$, purely kinetic.

Question 1.1.103 Statement: “A quantum state can be expanded in more than one valid orthonormal basis.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Any complete orthonormal basis can be used to expand the same physical state; the amplitudes just look different.

2-Mark Multiple Choice Questions (12 Questions for Unit 1)

Question 1.2.1 For $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$, the probability of measuring $|1\rangle$ is:

- A 3/5

- B $4/5$
C $16/25$
D $9/25$

Answer: C

SOLUTION (2 marks): Born rule: $p_1 = |c_1|^2 = (4/5)^2 = 16/25 = 0.64$. Check normalization: $9/25 + 16/25 = 1$.

Mark breakdown:

- 0.5 mark: Born rule statement
- 0.5 mark: Normalization check
- 1 mark: Correct value

Question 1.2.2 The expectation value $\langle \sigma_z \rangle$ for $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is:

- A 1
B -1
C 0
D $1/\sqrt{2}$

Answer: C

SOLUTION (2 marks): $\langle \sigma_z \rangle = |c_0|^2 - |c_1|^2 = \frac{1}{2} - \frac{1}{2} = 0$. The state $|+\rangle$ lies on the equator of the Bloch sphere, so its z -projection vanishes.

Mark breakdown:

- 1 mark: Formula $|c_0|^2 - |c_1|^2$
- 0.5 mark: Substitution
- 0.5 mark: Bloch-sphere interpretation

Question 1.2.3 A qubit sits at Bloch angles $\theta = \pi/3$, $\phi = 0$. The probability of measuring $|0\rangle$ is:

- A $1/2$
B $3/4$
C $1/4$
D $1/3$

Answer: B

SOLUTION (2 marks): $P(0) = \cos^2(\theta/2) = \cos^2(\pi/6) = (\sqrt{3}/2)^2 = 3/4$.

Mark breakdown:

- 1 mark: Half-angle parametrization
- 0.5 mark: Evaluation of $\cos(\pi/6)$
- 0.5 mark: Final value

Question 1.2.4 The Bloch vector of the state $\frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$ points along:

A $+\hat{x}$

B $-\hat{y}$

C $+\hat{z}$

D $-\hat{z}$

Answer: B

SOLUTION (2 marks): Compare with $\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$: here $\theta = \pi/2$ and $e^{i\phi} = -i$, so $\phi = -\pi/2$. The Bloch vector is $(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) = (0, -1, 0)$.

Mark breakdown:

- 0.75 mark: Extracting θ
- 0.75 mark: Extracting ϕ
- 0.5 mark: Vector identification

Question 1.2.5 Two of the Pauli operators satisfy $\sigma_x \sigma_y = ?$

A $-i\sigma_z$

B $i\sigma_z$

C σ_z

D I

Answer: B

SOLUTION (2 marks): Using $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$:

$$\sigma_x \sigma_y = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_z.$$

Generally $\sigma_j \sigma_k = \delta_{jk} I + i\epsilon_{jkl} \sigma_l$.

Mark breakdown:

- 1 mark: Matrix multiplication
- 0.5 mark: Identification with $i\sigma_z$
- 0.5 mark: General algebra statement

Question 1.2.6 An operator A has eigenvalues $\{+1, -1\}$ and $A = A^\dagger$, $A^2 = I$. Such an operator is:

- A Only Hermitian
- B Both Hermitian and unitary
- C Only unitary
- D Neither

Answer: B

SOLUTION (2 marks): $A^\dagger A = AA = A^2 = I$, so A is unitary as well as Hermitian. The Pauli matrices are exactly of this type, which is why they act both as observables and as gates.

Mark breakdown:

- 1 mark: Showing $A^\dagger A = I$
- 0.5 mark: Correct classification
- 0.5 mark: Example

Question 1.2.7 If $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is measured in the $\{|+\rangle, |-\rangle\}$ basis, the outcome is:

- A $|+\rangle$ with probability 1
- B $|-\rangle$ with probability 1
- C Either with probability 0.5
- D Undefined

Answer: A

SOLUTION (2 marks): The state is exactly $|+\rangle$, an eigenstate of the measured observable σ_x . Then $|\langle+|\psi\rangle|^2 = 1$ and $|\langle-|\psi\rangle|^2 = 0$. Randomness is basis-dependent, not intrinsic to the state.

Mark breakdown:

- 1 mark: Recognizing the eigenstate
- 0.5 mark: Overlap computation
- 0.5 mark: Interpretation

Question 1.2.8 For a Gaussian wavepacket the product $\Delta x \Delta p$ equals:

- A 0
- B $\hbar/2$ (minimum uncertainty)
- C $\hbar$
- D $2\hbar$

Answer: B

SOLUTION (2 marks): The Robertson relation gives $\Delta x \Delta p \geq \frac{1}{2} |\langle [\hat{x}, \hat{p}] \rangle| = \hbar/2$. Gaussian states saturate this bound and are therefore called minimum-uncertainty states.

Mark breakdown:

- 1 mark: Uncertainty relation
- 0.5 mark: Saturation condition
- 0.5 mark: Answer

Question 1.2.9 A system evolves under $H = \frac{\hbar\omega}{2}\sigma_z$ from $|+\rangle$. After time $t = \pi/\omega$ the state is (up to global phase):

- A $|+\rangle$
- B $|-\rangle$
- C $|0\rangle$
- D $|1\rangle$

Answer: B

SOLUTION (2 marks): $U(t) = e^{-i\omega t\sigma_z/2} = \text{diag}(e^{-i\omega t/2}, e^{+i\omega t/2})$. At $t = \pi/\omega$ this gives $\text{diag}(e^{-i\pi/2}, e^{i\pi/2})$, so

$$|+\rangle \rightarrow \frac{1}{\sqrt{2}}(e^{-i\pi/2}|0\rangle + e^{i\pi/2}|1\rangle) \equiv \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle.$$

This is Larmor precession by π about the z -axis.

Mark breakdown:

- 0.75 mark: Evolution operator
- 0.75 mark: Phase evaluation
- 0.5 mark: Identification and precession picture

Question 1.2.10 In a double-slit run, slit amplitudes are ψ_1 and ψ_2 with $|\psi_1| = |\psi_2| = a$ and relative phase δ . The screen intensity is:

- A $2a^2$
- B $2a^2(1 + \cos \delta)$
- C $a^2 \cos \delta$
- D $4a^2$ always

Answer: B

SOLUTION (2 marks): $|\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\text{Re}(\psi_1\psi_2^*) = 2a^2 + 2a^2 \cos \delta$. Maxima $4a^2$ at $\delta = 0$, minima 0 at $\delta = \pi$.

Mark breakdown:

- 0.75 mark: Expanding the modulus square
- 0.75 mark: Cross term
- 0.5 mark: Fringe extremes

Question 1.2.11 The inner product $\langle 0|+\rangle$ where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ equals:

- A 0
- B 1
- C $1/\sqrt{2}$
- D $1/2$

Answer: C

SOLUTION (2 marks): $\langle 0|+\rangle = \frac{1}{\sqrt{2}}(\langle 0|0\rangle + \langle 0|1\rangle) = \frac{1}{\sqrt{2}}(1 + 0) = \frac{1}{\sqrt{2}}$. Note the corresponding probability is $|1/\sqrt{2}|^2 = 1/2$; amplitude and probability must not be confused.

Mark breakdown:

- 1 mark: Using orthonormality
- 0.5 mark: Amplitude value
- 0.5 mark: Distinguishing amplitude from probability

Question 1.2.12 An electron is confined to a 1D infinite well of width $L = 1$ nm. The ground-state energy is approximately:

- A 0.038 eV
- B 0.376 eV
- C 1.13 eV
- D 3.76 eV

Answer: B

SOLUTION (2 marks): $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$. For $n = 1$, $L = 10^{-9}$ m:

$$E_1 = \frac{\pi^2(1.055 \times 10^{-34})^2}{2(9.11 \times 10^{-31})(10^{-18})} \approx 6.0 \times 10^{-20} \text{ J} \approx 0.376 \text{ eV}.$$

Mark breakdown:

- 0.75 mark: Correct formula
- 0.75 mark: Numerical substitution
- 0.5 mark: Conversion to eV

4-Mark Subjective Questions ($5 \times 4\text{M} = 20$ marks for Unit 1)

[NUMERICAL QUESTIONS: 3 per block]

Question 1.3.1 – NUMERICAL (4 marks)

State Normalization and Measurement Statistics

A qubit is prepared as

$$|\psi\rangle = N(2|0\rangle + (1+i)|1\rangle).$$

- (a) Determine the normalization constant N . (1.5 marks)

(b) Find the probabilities of measuring $|0\rangle$ and $|1\rangle$. (1 mark)

(c) Compute $\langle\sigma_z\rangle$. (1.5 marks)

(a) Normalization (1.5 marks)

$$\langle\psi|\psi\rangle = |N|^2(|2|^2 + |1+i|^2) = |N|^2(4+2) = 6|N|^2 = 1$$

$$N = \frac{1}{\sqrt{6}}$$

(b) Probabilities (1 mark)

$$p_0 = \frac{4}{6} = \frac{2}{3}, \quad p_1 = \frac{2}{6} = \frac{1}{3}$$

Check: $p_0 + p_1 = 1$.

(c) Expectation value (1.5 marks)

$$\langle\sigma_z\rangle = (+1)p_0 + (-1)p_1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \approx 0.333$$

The Bloch vector therefore has z -component $1/3$, so the state leans towards the north pole but is not an eigenstate of σ_z .

Question 1.3.2 – NUMERICAL (4 marks)

Bloch Sphere Representation

Consider

$$|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}e^{i\pi/3}|1\rangle.$$

(a) Extract the Bloch angles θ and ϕ . (1.5 marks)

(b) Compute the Bloch vector $\vec{r} = (r_x, r_y, r_z)$. (1.5 marks)

(c) Verify that the state is pure. (1 mark)

(a) Angles (1.5 marks)

Comparing with $\cos \frac{\theta}{2}|0\rangle + e^{i\phi} \sin \frac{\theta}{2}|1\rangle$:

$$\cos \frac{\theta}{2} = \frac{1}{2} \Rightarrow \frac{\theta}{2} = \frac{\pi}{3} \Rightarrow \theta = \frac{2\pi}{3}$$

$$\sin \frac{\theta}{2} = \frac{\sqrt{3}}{2} \checkmark, \quad \phi = \frac{\pi}{3}$$

(b) Bloch vector (1.5 marks)

$$r_x = \sin \theta \cos \phi = \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{\sqrt{3}}{4} \approx 0.433$$

$$r_y = \sin \theta \sin \phi = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{4} = 0.75$$

$$r_z = \cos \theta = -\frac{1}{2}$$

(c) Purity check (1 mark)

$$|\vec{r}|^2 = \frac{3}{16} + \frac{9}{16} + \frac{4}{16} = 1$$

The Bloch vector has unit length, so the state lies on the surface of the sphere and is pure, consistent with $\text{Tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2) = 1$.

Question 1.3.3 – NUMERICAL (4 marks)

Unitary Evolution of a Qubit

A qubit starts in $|0\rangle$ and evolves under $H = \frac{\hbar\Omega}{2}\sigma_x$.

- Write $U(t) = e^{-iHt/\hbar}$ in matrix form. (1.5 marks)
- Find $|\psi(t)\rangle$. (1 mark)
- Compute $P(1)$ at t and give the first time at which it equals 1. (1.5 marks)

(a) Evolution operator (1.5 marks)

Since $\sigma_x^2 = I$, the exponential resums to

$$U(t) = \cos\left(\frac{\Omega t}{2}\right) I - i \sin\left(\frac{\Omega t}{2}\right) \sigma_x = \begin{pmatrix} \cos \frac{\Omega t}{2} & -i \sin \frac{\Omega t}{2} \\ -i \sin \frac{\Omega t}{2} & \cos \frac{\Omega t}{2} \end{pmatrix}$$

(b) Evolved state (1 mark)

$$|\psi(t)\rangle = U(t)|0\rangle = \cos\left(\frac{\Omega t}{2}\right) |0\rangle - i \sin\left(\frac{\Omega t}{2}\right) |1\rangle$$

(c) Transition probability (1.5 marks)

$$P(1) = \left| -i \sin \frac{\Omega t}{2} \right|^2 = \sin^2\left(\frac{\Omega t}{2}\right)$$

This equals 1 when $\frac{\Omega t}{2} = \frac{\pi}{2}$, i.e.

$$t = \frac{\pi}{\Omega}.$$

These are Rabi oscillations of period $2\pi/\Omega$; the pulse at $t = \pi/\Omega$ is the π -pulse (an X gate up to phase).

Question 1.3.4 – THEORY (4 marks)

State the four basic postulates of quantum mechanics (state space, evolution, measurement, composite systems) and give one short mathematical statement for each.

Solution:**1. State space (1 mark)**

An isolated physical system is associated with a complex Hilbert space $\mathcal{H}$. The system is completely described by a unit vector $|\psi\rangle \in \mathcal{H}$, with $\langle\psi|\psi\rangle = 1$.

2. Evolution (1 mark)

Closed-system evolution is unitary:

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U = e^{-iHt/\hbar}, \quad i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle.$$

Unitarity preserves the norm and hence total probability.

3. Measurement (1 mark)

A measurement is described by a set $\{M_m\}$ with $\sum_m M_m^\dagger M_m = I$. Outcome m occurs with probability

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle,$$

and the post-measurement state is $M_m|\psi\rangle/\sqrt{p(m)}$. For projective measurement of a Hermitian A this reduces to the Born rule $p_a = |\langle a|\psi\rangle|^2$.

4. Composite systems (1 mark)

The state space of a composite system is the tensor product of the components:

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n.$$

For n qubits, $\dim \mathcal{H} = 2^n$. States that cannot be written as a single product $|\psi_1\rangle \otimes |\psi_2\rangle$ are entangled, which is the origin of the exponential resource in quantum computing.

Question 1.3.5 – SCENARIO NUMERICAL (4 marks)**Electron Double-Slit Interferometer**

Electrons of de Broglie wavelength $\lambda = 5$ nm pass through two slits separated by $d = 500$ nm. The screen is at $L = 1$ m.

- (a) Write the condition for constructive interference and compute the fringe spacing. (2 marks)
- (b) The electron rate is reduced so that only one electron is in the apparatus at a time. Explain what is observed and why. (1 mark)
- (c) A which-path detector of efficiency 100% is switched on. State the resulting pattern. (1 mark)

(a) Fringe spacing (2 marks)

Constructive interference requires a path difference of an integer number of wavelengths:

$$d \sin \theta = m\lambda, \quad m = 0, \pm 1, \pm 2, \dots$$

For small angles, $\sin \theta \approx y/L$, so the position of the m -th maximum is $y_m = m\lambda L/d$ and the fringe spacing is

$$\Delta y = \frac{\lambda L}{d} = \frac{(5 \times 10^{-9})(1)}{5 \times 10^{-7}} = 1 \times 10^{-2} \text{ m} = 10 \text{ mm}.$$

(b) Single-electron regime (1 mark)

Each electron is detected as one localized dot at a random position. As dots accumulate, the same interference pattern emerges. The interference is between the two path amplitudes of a single electron:

$$P(y) = |\psi_1(y) + \psi_2(y)|^2,$$

not between different electrons.

(c) With which-path detection (1 mark)

Path information makes the two branches orthogonal in the detector degree of freedom, so the cross term averages to zero:

$$P(y) = |\psi_1(y)|^2 + |\psi_2(y)|^2.$$

The fringes vanish and only the broad single-slit envelope (the classical two-band sum) remains. This is complementarity: full which-path knowledge implies zero fringe visibility.

UNIT 2: QUANTUM STATES ALGEBRA

1-Mark Multiple Choice Questions (20 Questions)

Question 2.1.1 A density matrix ρ must satisfy:

- A $\rho = \rho^\dagger, \rho \geq 0, \text{Tr}(\rho) = 1$
- B $\rho^2 = I$
- C $\text{Tr}(\rho) = 0$
- D ρ unitary

Answer: A

Solution (1 mark): Hermiticity gives real eigenvalues, positivity makes them non-negative probabilities, and unit trace normalizes them.

Question 2.1.2 The purity condition for a pure state is:

- A $\text{Tr}(\rho^2) < 1$
- B $\text{Tr}(\rho^2) = 1$ and $\rho^2 = \rho$
- C $\text{Tr}(\rho^2) = 2$
- D $\rho = I/2$

Answer: B

Solution (1 mark): A pure state is a rank-1 projector $\rho = |\psi\rangle\langle\psi|$, so it is idempotent and $\text{Tr}(\rho^2) = 1$.

Question 2.1.3 The maximally mixed state of a single qubit is:

- A $|0\rangle\langle 0|$
- B $I_2/2$
- C $|+\rangle\langle +|$
- D I_2

Answer: B

Solution (1 mark): Equal eigenvalues $1/2$ give maximum von Neumann entropy $S = 1$ bit and minimum purity $\text{Tr}(\rho^2) = 1/2$.

Question 2.1.4 Two different ensembles that give the same density matrix are:

- A Distinguishable by a clever measurement
- B Physically indistinguishable
- C Always both pure
- D Always orthogonal

Answer: B

Solution (1 mark): All measurement statistics follow from $\text{Tr}(\rho M)$. Identical ρ means identical predictions for every observable.

Question 2.1.5 The Bloch-vector representation of a qubit density matrix is:

- A $\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma})$
- B $\rho = \vec{r} \cdot \vec{\sigma}$
- C $\rho = I + \vec{r} \cdot \vec{\sigma}$
- D $\rho = \frac{1}{2}\vec{r} \cdot \vec{\sigma}$

Answer: A

Solution (1 mark): Pure states have $|\vec{r}| = 1$ (surface), mixed states $|\vec{r}| < 1$ (interior), and $\vec{r} = 0$ is the maximally mixed state at the centre.

Question 2.1.6 The set of density matrices forms:

- A A convex set whose extreme points are the pure states
- B A vector space
- C A group under multiplication
- D A discrete set

Answer: A

Solution (1 mark): Any convex combination $\sum_i p_i \rho_i$ is again a valid state. Pure states cannot be written as mixtures of others, so they are the extreme points.

Question 2.1.7 The essential difference between a Cbit and a Qbit is that a Qbit:

- A Stores more classical numbers

B Admits coherent superposition with relative phase

C Is always faster

D Cannot be measured

Answer: B

Solution (1 mark): A Cbit is a probability distribution over $\{0,1\}$; a Qbit is a complex amplitude vector where the relative phase enables interference.

Question 2.1.8 The EPR argument was originally intended to show that quantum mechanics is:

A Incorrect

B Incomplete

C Non-linear

D Deterministic

Answer: B

Solution (1 mark): Einstein, Podolsky and Rosen argued that assuming locality and realism, the wavefunction cannot be a complete description, suggesting hidden variables.

Question 2.1.9 The von Neumann entropy is defined as:

A $\text{Tr}(\rho^2)$

B $-\text{Tr}(\rho \log_2 \rho)$

C $-\text{Tr}(\rho)$

D $\log_2 \text{Tr}(\rho)$

Answer: B

Solution (1 mark): It is the quantum analogue of the Shannon entropy, computed from the eigenvalues: $S = -\sum_i \lambda_i \log_2 \lambda_i$.

Question 2.1.10 A reduced density matrix is obtained by:

A Taking the determinant

B Partial trace over the unobserved subsystem

C Transposing

D Normalizing the state vector

Answer: B

Solution (1 mark): $\rho_A = \text{Tr}_B(\rho_{AB})$ gives all statistics accessible to a party holding only subsystem A .

Question 2.1.11 Non-orthogonal quantum states are:

A Perfectly distinguishable in one shot

B Not perfectly distinguishable

C Always identical

D Always entangled

Answer: B

Solution (1 mark): Perfect discrimination requires orthogonal projectors. If $\langle\psi_1|\psi_2\rangle \neq 0$ there is an unavoidable error probability, which underlies the security of QKD.

Question 2.1.12 The Hadamard gate matrix is:

A $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

B $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

C $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

D $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

Answer: A

Solution (1 mark): $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, and $H^2 = I$. It maps between the Z and X bases.

Question 2.1.13 Every quantum logic gate must be:

A Irreversible

- B Unitary, hence reversible
- C Hermitian
- D Diagonal

Answer: B

Solution (1 mark): $U^\dagger U = I$ means $U^{-1} = U^\dagger$ always exists. Classical AND and OR are irreversible and therefore have no direct unitary counterpart.

Question 2.1.14 The CNOT gate acting on $|10\rangle$ (first qubit control) gives:

- A $|10\rangle$
- B $|11\rangle$
- C $|00\rangle$
- D $|01\rangle$

Answer: B

Solution (1 mark): Control is 1, so the target flips: $|1\rangle|0\rangle \rightarrow |1\rangle|1\rangle$. In general $|a, b\rangle \rightarrow |a, a \oplus b\rangle$.

Question 2.1.15 The Toffoli (CCNOT) gate is important because it is:

- A The only single-qubit gate
- B Universal for reversible classical computation
- C Non-unitary
- D Equivalent to a Hadamard

Answer: B

Solution (1 mark): Toffoli can simulate NAND (set the target to $|1\rangle$), so any classical circuit can be embedded reversibly into a quantum circuit.

Question 2.1.16 A universal gate set for quantum computation is:

- A CNOT alone
- B Arbitrary single-qubit rotations plus CNOT
- C Hadamard alone

D SWAP plus measurement

Answer: B

Solution (1 mark): Any n -qubit unitary can be decomposed into single-qubit rotations and CNOTs. A common discrete set is $\{H, T, \text{CNOT}\}$.

Question 2.1.17 Applying the phase gate $S = \text{diag}(1, i)$ to $|+\rangle$ produces:

A $|+\rangle$

B $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$

C $|-\rangle$

D $|1\rangle$

Answer: B

Solution (1 mark): S multiplies the $|1\rangle$ amplitude by i , rotating the Bloch vector by 90° about the z -axis: $|+\rangle \rightarrow |+i\rangle$.

Question 2.1.18 In Qiskit, the standard Bell-pair circuit is:

A `qc.x(0); qc.z(1)`

B `qc.h(0); qc.cx(0,1)`

C `qc.swap(0,1)`

D `qc.measure_all()`

Answer: B

Solution (1 mark): Hadamard creates the superposition and CNOT correlates the qubits, producing $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

Question 2.1.19 Running a Qiskit circuit on the `qasm_simulator` with `shots=1024` returns:

A The exact statevector

B A histogram of measured bitstring counts

C The Hamiltonian

D The unitary matrix

Answer: B

Solution (1 mark): The QASM simulator samples the output distribution, so counts approach $N \cdot p_i$ with statistical fluctuation $\sim \sqrt{N}$. The statevector simulator is used for exact amplitudes.

Question 2.1.20 Identical bosons in the same mode are described by a wavefunction that is:

- A Antisymmetric under exchange
- B Symmetric under exchange
- C Zero
- D Arbitrary

Answer: B

Solution (1 mark): Bosons take the symmetric combination and can share a state; fermions take the antisymmetric one, which vanishes for identical labels (Pauli exclusion).

True/False Style Questions (20 Questions)

Question 2.1.21 Statement: “Every density matrix ρ must satisfy $\text{Tr}(\rho) = 1$.” This statement is:

- A True
- B False
- C True only for pure states
- D True only for 2-qubit systems

Answer: A

Solution (1 mark): True. Unit trace normalizes the eigenvalues of ρ so they form a valid probability distribution.

Question 2.1.22 Statement: “A pure state satisfies $\rho^2 = \rho$.” This statement is:

- A True
- B False
- C True only for maximally mixed states
- D True only for entangled states

Answer: A

Solution (1 mark): True. A pure state is a rank-1 projector, $\rho = |\psi\rangle\langle\psi|$, and projectors are idempotent.

Question 2.1.23 Statement: “The maximally mixed qubit state $I/2$ has $\text{Tr}(\rho^2) = 1$.” This statement is:

- A True
- B False
- C True only in the Bell basis
- D True only for a 3-qubit system

Answer: B

Solution (1 mark): False. $\text{Tr}((I/2)^2) = 1/2$; purity equals 1 only for pure states.

Question 2.1.24 Statement: “Two different physical ensembles can give rise to exactly the same density matrix.” This statement is:

- A True
- B False
- C True only for pure ensembles
- D True only for classical mixtures

Answer: A

Solution (1 mark): True. E.g., an equal mixture of $|0\rangle, |1\rangle$ and an equal mixture of $|+\rangle, |-\rangle$ both give $I/2$.

Question 2.1.25 Statement: “All quantum logic gates must be unitary.” This statement is:

- A True
- B False
- C True only for single-qubit gates
- D True only for measurement gates

Answer: A

Solution (1 mark): True. Gates must preserve the norm and be reversible, i.e. $U^\dagger U = I$.

Question 2.1.26 Statement: “The CNOT gate is its own inverse.” This statement is:

- A True
- B False
- C True only when the control is $|0\rangle$
- D True only in Qiskit

Answer: A

Solution (1 mark): True. $\text{CNOT}^2 = I_4$, since applying the same conditional flip twice restores the original state.

Question 2.1.27 Statement: “Classical AND and OR gates are reversible.” This statement is:

- A True
- B False
- C True only for XOR
- D True only in Boolean algebra

Answer: B

Solution (1 mark): False. They lose information (two inputs map to one output), so they have no direct unitary counterpart.

Question 2.1.28 Statement: “The Toffoli gate is universal for reversible classical computation.” This statement is:

- A True
- B False
- C True only for 2-bit circuits
- D True only when combined with SWAP

Answer: A

Solution (1 mark): True. Setting the target qubit to $|1\rangle$, the Toffoli gate implements NAND, from which any classical circuit can be built reversibly.

Question 2.1.29 Statement: “The Hadamard gate satisfies $H^2 = I$.” This statement is:

- A True
- B False
- C True only up to global phase
- D True only for $|0\rangle$

Answer: A

Solution (1 mark): True. H is Hermitian and unitary, so it is its own inverse.

Question 2.1.30 Statement: “Non-orthogonal quantum states can always be perfectly distinguished in a single measurement.” This statement is:

- A True
- B False
- C True only for mixed states
- D True only for macroscopic systems

Answer: B

Solution (1 mark): False. Non-zero overlap means no measurement can discriminate them with certainty; this is the basis of QKD security.

Question 2.1.31 Statement: “The EPR paper argued that quantum mechanics is incomplete.” This statement is:

- A True
- B False
- C True only after Bell’s theorem
- D True only for mixed states

Answer: A

Solution (1 mark): True. Einstein, Podolsky and Rosen argued that locality plus realism implies the wavefunction cannot be a complete description.

Question 2.1.32 Statement: “The von Neumann entropy of a pure state is zero.” This statement is:

- A True
- B False
- C True only for qubits
- D True only for the maximally mixed state

Answer: A

Solution (1 mark): True. A pure state has eigenvalues $\{1, 0, 0, \dots\}$, giving $S = -1 \log_2 1 = 0$.

Question 2.1.33 Statement: “A reduced density matrix is obtained by taking the determinant of the full state.” This statement is:

- A True
- B False
- C True only for pure global states
- D True only for 2-qubit systems

Answer: B

Solution (1 mark): False. It is obtained by a *partial trace* over the unobserved subsystem, not a determinant.

Question 2.1.34 Statement: “The set of all valid density matrices forms a convex set.” This statement is:

- A True
- B False
- C True only for qubits
- D True only for unitary operators

Answer: A

Solution (1 mark): True. Any convex combination $\sum_i p_i \rho_i$ is again a valid state, and pure states are its extreme points.

Question 2.1.35 Statement: “In Qiskit, a fresh `QuantumCircuit` initializes qubits into $|1\rangle$ by default.” This statement is:

- A True
- B False
- C True only after a reset
- D True only on real hardware

Answer: B

Solution (1 mark): False. Qubits always initialize into $|0\rangle$; an X gate is needed to flip to $|1\rangle$.

Question 2.1.36 Statement: “Identical fermions can occupy exactly the same quantum state.” This statement is:

- A True
- B False
- C True only at low temperature
- D True only for entangled fermions

Answer: B

Solution (1 mark): False. The antisymmetric wavefunction vanishes for identical labels, which is the Pauli exclusion principle.

Question 2.1.37 Statement: “A qubit stores strictly more classical information than a single bit because of superposition.” This statement is:

- A True
- B False
- C True only after measurement
- D True only for entangled qubits

Answer: B

Solution (1 mark): False. A single projective measurement of one qubit still yields only one classical bit of information; superposition is not extra classical storage.

Question 2.1.38 Statement: “The Bloch vector of $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$ is the zero vector.” This statement is:

- A True
- B False
- C True only in the X basis
- D Cannot be determined

Answer: A

Solution (1 mark): True. This equal mixture is exactly $I/2$, which sits at the centre of the Bloch sphere.

Question 2.1.39 Statement: “A universal gate set requires arbitrary single-qubit rotations plus a single entangling gate such as CNOT.” This statement is:

- A True
- B False
- C True only for 2-qubit circuits
- D True only in Qiskit

Answer: A

Solution (1 mark): True. Any n -qubit unitary can be decomposed into single-qubit rotations and CNOTs.

Question 2.1.40 Statement: “Applying a unitary gate to the maximally mixed state always changes it.” This statement is:

- A True
- B False
- C True only for two-qubit unitaries
- D True only for non-Clifford gates

Answer: B

Solution (1 mark): False. $U(I/2)U^\dagger = I/2$ for any unitary U ; the maximally mixed state is invariant under all unitaries.

Additional Easy Recall Questions (63 Questions)

Question 2.1.41 Statement: “A density matrix with $\text{Tr}(\rho^2) < 1$ describes a mixed state.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Purity below 1 is precisely the signature of a mixed (non-pure) quantum state.

Question 2.1.42 Statement: “The Pauli X gate acts as a classical bit-flip on a qubit.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$, exactly like a classical NOT gate.

Question 2.1.43 Statement: “The Pauli Z gate flips the sign of the $|1\rangle$ amplitude while leaving $|0\rangle$ unchanged.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$; this is called a phase flip.

Question 2.1.44 Statement: “The S gate is defined as $\text{diag}(1, i)$, sometimes called the $\sqrt{Z}$ gate.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Applying S twice gives Z , since $i^2 = -1$, matching $\text{diag}(1, -1)$.

Question 2.1.45 Statement: “The T gate applies a $\pi/8$ phase rotation and satisfies $T^2 = S$.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. $T = \text{diag}(1, e^{i\pi/4})$, and squaring it gives $\text{diag}(1, i) = S$.

Question 2.1.46 Statement: “The SWAP gate exchanges the states of two qubits.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. $\text{SWAP}|ab\rangle = |ba\rangle$ for any two qubits a, b .

Question 2.1.47 Statement: “Measurement of a quantum state is a unitary, fully reversible process.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: B

Solution (1 mark): False. Measurement is not unitary; it is a probabilistic, irreversible collapse of the state.

Question 2.1.48 Statement: “The diagonal entries of a density matrix in a given basis represent the probabilities of measuring each basis outcome.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This follows directly from the Born rule applied to ρ .

Question 2.1.49 Statement: “Off-diagonal entries of a density matrix are sometimes called quantum coherences.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. They encode the phase relationships between basis states, which are lost under decoherence.

Question 2.1.50 Statement: “Decoherence tends to suppress the off-diagonal entries of a density matrix over time.” This statement is:

- A True

- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Environmental interaction randomizes relative phases, driving the off-diagonal (coherence) terms toward zero.

Question 2.1.51 Statement: “The partial trace is a linear operation on density matrices.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Partial trace is defined as a sum over basis elements of one subsystem, which is manifestly linear.

Question 2.1.52 Statement: “A product state of two qubits has zero entanglement entropy.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. For $\rho_{AB} = \rho_A \otimes \rho_B$, the reduced state ρ_A is unchanged and its entropy reflects no correlation with B .

Question 2.1.53 Statement: “A maximally entangled two-qubit state has a reduced single-qubit entropy of 1 bit.” This statement is:

- A True
- B False
- C True only for 2-qubit systems

D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Its reduced state is the maximally mixed state $I/2$, which has entropy $S = 1$ bit.

Question 2.1.54 Statement: “Applying the identity gate to a qubit leaves its state completely unchanged.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. By definition, $I|\psi\rangle = |\psi\rangle$.

Question 2.1.55 Statement: “ $X^2 = I$ for the Pauli X gate.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Applying a bit-flip twice restores the original state.

Question 2.1.56 Statement: “ $Z^2 = I$ for the Pauli Z gate.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Applying a phase-flip twice restores the original sign.

Question 2.1.57 Statement: “ $Y^2 = I$ for the Pauli Y gate.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Like X and Z , the Pauli Y operator is also its own inverse.

Question 2.1.58 Statement: “The identity $HXH = Z$ holds for the Hadamard gate conjugating X .” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Conjugation by Hadamard swaps the roles of the X and Z bases, a key identity in quantum circuit design.

Question 2.1.59 Statement: “The identity $HZH = X$ holds for the Hadamard gate conjugating Z .” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This is the reverse of the $HXH = Z$ identity, reflecting Hadamard’s role in swapping bases.

Question 2.1.60 Statement: “In a CNOT gate, the target qubit flips only when the control qubit is in state $|1\rangle$.” This statement is:

- A True

- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is the defining conditional-flip behaviour of CNOT.

Question 2.1.61 Statement: “In a CNOT gate, the control qubit’s state can be changed by the operation.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: B

Solution (1 mark): False. The control qubit is always left unchanged; only the target may flip.

Question 2.1.62 Statement: “The SWAP gate can be constructed from three CNOT gates.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Three CNOTs applied in an alternating control/target pattern implement a SWAP.

Question 2.1.63 Statement: “A single-qubit gate alone can create entanglement between two qubits.” This statement is:

- A True
- B False
- C True only in the classical limit

D False except for entangled states

Answer: B

Solution (1 mark): False. Entangling two qubits requires an interaction (multi-qubit) gate such as CNOT; single-qubit gates act independently.

Question 2.1.64 Statement: “A two-qubit gate is generally required to entangle two previously unentangled qubits.” This statement is:

A True

B False

C True only for qubits

D False only after measurement

Answer: A

Solution (1 mark): True. This is why gates like CNOT or CZ are essential ingredients of any universal quantum computer.

Question 2.1.65 Statement: “Measurement outcome probabilities for a quantum state must always sum to exactly 1.” This statement is:

A True

B False

C True only for pure states

D False only for mixed states

Answer: A

Solution (1 mark): True. This follows directly from normalization of the state (or unit trace of the density matrix).

Question 2.1.66 Statement: “The Born rule generalizes to mixed states as $p(m) = \text{Tr}(\rho M_m)$.” This statement is:

A True

B False

C True only in Qiskit

D False only on real hardware

Answer: A

Solution (1 mark): True. This reduces to the usual $|\langle a|\psi\rangle|^2$ formula when ρ is pure.

Question 2.1.67 Statement: “A diagonal density matrix in some basis behaves exactly like a classical probability distribution in that basis.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. With no off-diagonal coherences, the state carries no quantum interference effects in that basis.

Question 2.1.68 Statement: “A Bell state can always be written as a simple tensor product of two single-qubit states.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: B

Solution (1 mark): False. Bell states are entangled and, by definition, cannot be factored into a product of two single-qubit states.

Question 2.1.69 Statement: “A separable two-qubit state can always be written as a tensor product, or a mixture of tensor products.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This is precisely the definition of a separable (non-entangled) state.

Question 2.1.70 Statement: “Quantum teleportation physically transports the original qubit itself across space.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: B

Solution (1 mark): False. Only the quantum information is transferred; the original qubit’s state is destroyed by measurement, and a new qubit at the receiver takes on that state.

Question 2.1.71 Statement: “The Toffoli gate requires exactly 3 qubits to define its action.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. It has two control qubits and one target qubit, for a total of three.

Question 2.1.72 Statement: “The Fredkin gate is a controlled-SWAP operation.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. It swaps two target qubits conditioned on the state of a third control qubit.

Question 2.1.73 Statement: “Universal quantum computation requires both single-qubit gates and at least one entangling two-qubit gate.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Single-qubit gates alone cannot generate entanglement, so a two-qubit entangling gate is also needed for universality.

Question 2.1.74 Statement: “The Clifford group is generated by the gates H , S , and CNOT.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. These three gates generate the Clifford group, though it alone is not universal for quantum computation.

Question 2.1.75 Statement: “The T gate is part of the Clifford group.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: B

Solution (1 mark): False. The T gate is a non-Clifford gate; adding it to the Clifford set makes the gate set universal.

Question 2.1.76 Statement: “The matrix form of the Pauli X gate is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This matrix swaps the amplitudes of $|0\rangle$ and $|1\rangle$, implementing a bit-flip.

Question 2.1.77 Statement: “The matrix form of the Pauli Z gate is $\text{diag}(1, -1)$.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. This leaves $|0\rangle$ unchanged and flips the sign of $|1\rangle$.

Question 2.1.78 Statement: “Every valid density matrix has trace exactly equal to 1.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Unit trace ensures the total probability across all outcomes sums to one.

Question 2.1.79 Statement: “The eigenvalues of any valid density matrix always lie between 0 and 1.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This follows from positivity ($\rho \geq 0$) combined with unit trace.

Question 2.1.80 Statement: “The eigenvalues of a valid density matrix always sum to 1.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This is just a restatement of the unit-trace condition in the eigenbasis.

Question 2.1.81 Statement: “A valid density matrix must be positive semi-definite.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This ensures all measurement probabilities computed from it are non-negative.

Question 2.1.82 Statement: “Entanglement between two qubits can be created using only local operations and classical communication (LOCC).” This statement is:

- A True
- B False

C True only above room temperature

D False only at zero temperature

Answer: B

Solution (1 mark): False. LOCC cannot create entanglement from a product state; genuine quantum interaction (an entangling gate) is required.

Question 2.1.83 Statement: “Local unitary operations applied to each subsystem separately cannot change the amount of entanglement between them.” This statement is:

A True

B False

C Cannot be determined

D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Local unitaries only rotate each party’s local basis; they leave the Schmidt coefficients, and hence the entanglement, unchanged.

Question 2.1.84 Statement: “The purity of any single-qubit state lies between 0.5 and 1.” This statement is:

A True

B False

C True only in the classical limit

D False except for entangled states

Answer: A

Solution (1 mark): True. Purity ranges from $1/2$ (maximally mixed) up to 1 (pure) for a single qubit.

Question 2.1.85 Statement: “A mixed quantum state always has purity strictly less than 1.” This statement is:

A True

B False

C True only for qubits

D False only after measurement

Answer: A

Solution (1 mark): True. $\text{Tr}(\rho^2) = 1$ if and only if the state is pure.

Question 2.1.86 Statement: “The von Neumann entropy of any pure state is always exactly zero.” This statement is:

A True

B False

C True only for pure states

D False only for mixed states

Answer: A

Solution (1 mark): True. A pure state has a single non-zero eigenvalue equal to 1, giving zero entropy.

Question 2.1.87 Statement: “The maximally mixed single-qubit state has the maximum possible von Neumann entropy of 1 bit.” This statement is:

A True

B False

C True only in Qiskit

D False only on real hardware

Answer: A

Solution (1 mark): True. Equal eigenvalues $1/2, 1/2$ maximize the entropy for a two-level system.

Question 2.1.88 Statement: “A Bell pair is also commonly referred to as an EPR pair.” This statement is:

A True

B False

C True only for 2-qubit systems

D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Both terms refer to the same maximally entangled two-qubit state, named after Einstein-Podolsky-Rosen and John Bell respectively.

Question 2.1.89 Statement: “In Qiskit, the syntax `qc.cx(0,1)` applies a CNOT gate with qubit 0 as control and qubit 1 as target.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This is the standard Qiskit convention for specifying control and target qubits in a CNOT.

Question 2.1.90 Statement: “In Qiskit, the syntax `qc.h(0)` applies a Hadamard gate to qubit 0.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This is the standard Qiskit method call for a single-qubit Hadamard gate.

Question 2.1.91 Statement: “Running more shots on a Qiskit simulator changes the true underlying probabilities of a quantum circuit.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: B

Solution (1 mark): False. More shots only improve the statistical estimate of the fixed underlying probabilities; the probabilities themselves are unchanged.

Question 2.1.92 Statement: “A statevector simulator returns the exact quantum amplitudes rather than sampled measurement counts.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Unlike a shot-based simulator, the statevector simulator directly reports the full complex amplitude vector.

Question 2.1.93 Statement: “Any single-qubit unitary can be built from a sequence of rotation gates around different axes.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. For example, any single-qubit unitary can be decomposed into R_z, R_y, R_z rotations (the Euler decomposition).

Question 2.1.94 Statement: “A classical NAND gate has a direct one-to-one unitary quantum equivalent with no extra qubits.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: B

Solution (1 mark): False. NAND is irreversible (two inputs map to one output), so it has no direct unitary equivalent without adding ancilla qubits, as in the Toffoli gate.

Question 2.1.95 Statement: “The reduced density matrix of a subsystem is obtained using the partial trace over the remaining subsystem(s).” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is the standard definition: $\rho_A = \text{Tr}_B(\rho_{AB})$.

Question 2.1.96 Statement: “Applying a unitary gate to a pure state always keeps it a pure state.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Unitary evolution preserves purity: $(U\rho U^\dagger)^2 = U\rho^2 U^\dagger$ when $\rho^2 = \rho$.

Question 2.1.97 Statement: “A quantum gate that is not unitary can still be directly implemented as a basic quantum logic gate.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: B

Solution (1 mark): False. All basic quantum logic gates must be unitary to preserve normalization and be reversible.

Question 2.1.98 Statement: “The number of distinct 1-qubit Pauli gates (excluding identity) is exactly 3: X , Y , Z .” This statement is:

- A True

- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. These are the three non-trivial Pauli operators.

Question 2.1.99 Statement: “A density matrix can be used to describe both pure and mixed quantum states within a single framework.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This generality is exactly why density matrices are used instead of state vectors when mixedness is possible.

Question 2.1.100 Statement: “The maximally mixed state of a single qubit corresponds to the center of the Bloch sphere.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. It has Bloch vector $\vec{r} = 0$, placing it at the sphere's center.

Question 2.1.101 Statement: “A CZ (controlled-Z) gate is another common two-qubit entangling gate besides CNOT.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. CZ applies a phase flip to the target only when the control is $|1\rangle$, and is also entangling.

Question 2.1.102 Statement: “Quantum gates can, in principle, always be decomposed into a sequence of single- and two-qubit gates.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is a foundational result: any multi-qubit unitary can be built from single-qubit rotations and two-qubit entangling gates.

Question 2.1.103 Statement: “A Bell state remains a Bell state after applying the same local unitary to both qubits, up to change of Bell basis.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Local unitaries preserve the entanglement structure, mapping Bell states to (possibly different) Bell states.

2-Mark Multiple Choice Questions (12 Questions for Unit 2)

Question 2.2.1 For $\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|$, the purity $\text{Tr}(\rho^2)$ is:

- A 1
- B 0.625
- C 0.5
- D 0.75

Answer: B

SOLUTION (2 marks): ρ is diagonal, so $\text{Tr}(\rho^2) = \sum_i \lambda_i^2 = (3/4)^2 + (1/4)^2 = 9/16 + 1/16 = 10/16 = 0.625$. Since $0.5 < 0.625 < 1$ the state is mixed but not maximally mixed.

Mark breakdown:

- 0.5 mark: Purity formula
- 1 mark: Computation
- 0.5 mark: Interpretation

Question 2.2.2 The von Neumann entropy of $\rho = \text{diag}(3/4, 1/4)$ is approximately:

- A 0 bits
- B 0.811 bits
- C 1 bit
- D 2 bits

Answer: B

SOLUTION (2 marks):

$$S = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = -0.75(-0.415) - 0.25(-2) = 0.311 + 0.5 = 0.811 \text{ bits.}$$

This is below the maximum of 1 bit, consistent with the purity found above.

Mark breakdown:

- 0.5 mark: Entropy formula
- 1 mark: Logarithm evaluation
- 0.5 mark: Final value

Question 2.2.3 The Bloch vector of $\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|$ is:

- A (0, 0, 1)
- B (0, 0, 1/2)
- C (1/2, 0, 0)
- D (0, 0, 0)

Answer: B

SOLUTION (2 marks): $r_i = \text{Tr}(\rho\sigma_i)$. Off-diagonal entries vanish so $r_x = r_y = 0$, while

$$r_z = \text{Tr}(\rho\sigma_z) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}.$$

$|\vec{r}| = 0.5 < 1$ places the state inside the sphere, i.e. mixed.

Mark breakdown:

- 0.75 mark: $r_i = \text{Tr}(\rho\sigma_i)$
- 0.75 mark: Component evaluation
- 0.5 mark: Mixedness conclusion

Question 2.2.4 The state $\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +|$ is:

- A Pure
- B Mixed with $|\vec{r}| = 1/\sqrt{2}$
- C Maximally mixed
- D Not a valid state

Answer: B

SOLUTION (2 marks): Bloch vectors of $|0\rangle$ and $|+\rangle$ are $(0, 0, 1)$ and $(1, 0, 0)$. Since the map $\rho \mapsto \vec{r}$ is linear,

$$\vec{r} = \frac{1}{2}(0, 0, 1) + \frac{1}{2}(1, 0, 0) = \left(\frac{1}{2}, 0, \frac{1}{2}\right), \quad |\vec{r}| = \frac{1}{\sqrt{2}} \approx 0.707.$$

Mixed, since $|\vec{r}| < 1$.

Mark breakdown:

- 0.75 mark: Component Bloch vectors
- 0.75 mark: Linear averaging
- 0.5 mark: Magnitude and conclusion

Question 2.2.5 Applying H to the maximally mixed state $I/2$ gives:

- A $|+\rangle\langle +|$
- B $I/2$ (unchanged)
- C $|0\rangle\langle 0|$
- D A non-physical state

Answer: B

SOLUTION (2 marks): $H \frac{I}{2} H^\dagger = \frac{1}{2} H H^\dagger = \frac{I}{2}$. The maximally mixed state is invariant under every unitary; it carries no information and sits at the centre of the Bloch sphere, which no rotation moves.

Mark breakdown:

- 1 mark: Conjugation calculation
- 0.5 mark: Unitary invariance
- 0.5 mark: Geometric argument

Question 2.2.6 Two non-orthogonal states with $|\langle\psi_1|\psi_2\rangle| = 1/\sqrt{2}$ can be distinguished with maximum success probability (equal priors):

- A 1
- B 0.854
- C 0.5
- D 0.707

Answer: B

SOLUTION (2 marks): The Helstrom bound for two pure states with overlap $|\langle\psi_1|\psi_2\rangle| = s$ is

$$P_{\max} = \frac{1}{2} \left(1 + \sqrt{1 - s^2} \right).$$

With $s = 1/\sqrt{2}$: $P_{\max} = \frac{1}{2}(1 + \sqrt{1/2}) = \frac{1}{2}(1.7071) \approx 0.854$.

Mark breakdown:

- 1 mark: Helstrom formula
- 0.5 mark: Substitution
- 0.5 mark: Final value

Question 2.2.7 For the singlet $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, the reduced state of qubit A is:

- A $|0\rangle\langle 0|$
- B $I_2/2$
- C $|+\rangle\langle +|$
- D Pure but unknown

Answer: B

SOLUTION (2 marks): Tracing over B:

$$\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I_2}{2}.$$

A globally pure state with a maximally mixed marginal is the signature of maximal entanglement: all the information lies in the correlations, none locally.

Mark breakdown:

- 1 mark: Partial trace
- 0.5 mark: Result
- 0.5 mark: Entanglement interpretation

Question 2.2.8 A CNOT applied to $|+\rangle|0\rangle$ produces a state that is:

- A Separable
- B Maximally entangled
- C Maximally mixed
- D Unchanged

Answer: B

SOLUTION (2 marks): $|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle.$

The reduced state of either qubit becomes $I/2$, so the entanglement entropy is 1 ebit.

Mark breakdown:

- 0.75 mark: Expansion of the input
- 0.75 mark: Gate action
- 0.5 mark: Entanglement identification

Question 2.2.9 In the EPR/Bohm setup with a singlet, if Alice measures σ_z and gets +1, Bob's outcome for σ_z is:

- A +1
- B -1
- C Random
- D Zero

Answer: B

SOLUTION (2 marks): The singlet contains only $|01\rangle$ and $|10\rangle$, so the outcomes are perfectly anti-correlated in every basis. Bob's marginal is still $I/2$ before he learns Alice's result, so no signal is transmitted.

Mark breakdown:

- 1 mark: Anti-correlation
- 0.5 mark: Basis independence
- 0.5 mark: No-signalling note

Question 2.2.10 The circuit H on qubit 0 followed by CNOT(0,1) followed by H on both qubits, acting on $|00\rangle$, produces a state that is:

- A Separable
- B Still maximally entangled
- C Mixed
- D $|00\rangle$

Answer: B

SOLUTION (2 marks): $(H \otimes H)|\Phi^+\rangle = |\Phi^+\rangle$. More generally, local unitaries cannot create or destroy entanglement, since the Schmidt coefficients are invariant under $U_A \otimes U_B$.

Mark breakdown:

- 0.75 mark: Circuit evaluation
- 0.75 mark: Local-unitary invariance
- 0.5 mark: Schmidt argument

Question 2.2.11 A two-qubit state $\rho_{AB} = \rho_A \otimes \rho_B$ has mutual information $I(A : B)$ equal to:

- A 0
- B 1 bit
- C 2 bits
- D Depends on ρ_A

Answer: A

SOLUTION (2 marks): For product states entropy is additive, $S_{AB} = S_A + S_B$, so

$$I(A : B) = S_A + S_B - S_{AB} = 0.$$

Zero mutual information means no correlations of any kind, classical or quantum.

Mark breakdown:

- 0.75 mark: Additivity for product states
- 0.75 mark: Substitution
- 0.5 mark: Interpretation

Question 2.2.12 A Qiskit run of the Bell circuit with 1000 shots on an ideal simulator gives counts closest to:

- A {'00': 1000}
- B {'00': 500, '11': 500}
- C {'00': 250, '01': 250, '10': 250, '11': 250}
- D {'01': 500, '10': 500}

Answer: B

SOLUTION (2 marks): $|\Phi^+\rangle$ gives $p_{00} = p_{11} = 0.5$ and $p_{01} = p_{10} = 0$. With 1000 shots the counts fluctuate about 500 each by roughly $\sqrt{1000}/2 \approx 16$. On real hardware small '01' and '10' counts appear from gate and readout error.

Mark breakdown:

- 1 mark: Correct probabilities
- 0.5 mark: Shot-noise estimate
- 0.5 mark: Hardware-noise remark

4-Mark Subjective Questions ($5 \times 4\text{M} = 20$ marks for Unit 2)

[NUMERICAL QUESTIONS: 3 per block]

Question 2.3.1 – NUMERICAL (4 marks)

Pure versus Mixed States

An ensemble contains $|0\rangle$ with probability 0.6 and $|+\rangle$ with probability 0.4.

- (a) Construct the density matrix ρ . (1.5 marks)
- (b) Compute the purity $\text{Tr}(\rho^2)$. (1.5 marks)
- (c) State whether the ensemble is pure or mixed and justify. (1 mark)

(a) Density matrix (1.5 marks)

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad |+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\rho = 0.6 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 0.4 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.2 \end{pmatrix}$$

Check: $\text{Tr}(\rho) = 0.8 + 0.2 = 1$.

(b) Purity (1.5 marks)

$$\rho^2 = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.2 \end{pmatrix} \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.2 \end{pmatrix} = \begin{pmatrix} 0.68 & 0.20 \\ 0.20 & 0.08 \end{pmatrix}$$

$$\text{Tr}(\rho^2) = 0.68 + 0.08 = 0.76$$

(c) Classification (1 mark)

Since $0.5 \leq 0.76 < 1$, the state is **mixed** but not maximally mixed. Equivalently the Bloch vector is $\vec{r} = (0.4, 0, 0.6)$ with $|\vec{r}| = \sqrt{0.52} \approx 0.721 < 1$, placing it strictly inside the Bloch sphere. Consistency check: $\text{Tr}(\rho^2) = \frac{1}{2}(1 + |\vec{r}|^2) = \frac{1}{2}(1.52) = 0.76$.

Question 2.3.2 – NUMERICAL (4 marks)
Ensemble Ambiguity of the Density Matrix

- (a) Show that the equal mixture of $|0\rangle$ and $|1\rangle$ gives $\rho_1 = I/2$. (1 mark)
- (b) Show that the equal mixture of $|+\rangle$ and $|-\rangle$ gives $\rho_2 = I/2$. (1.5 marks)
- (c) Explain the physical significance of $\rho_1 = \rho_2$. (1.5 marks)

(a) Computational-basis mixture (1 mark)

$$\rho_1 = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}$$

(b) Hadamard-basis mixture (1.5 marks)

$$|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad |-\rangle\langle -| = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\rho_2 = \frac{1}{2} \left(\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}$$

The off-diagonal terms cancel exactly.

(c) Significance (1.5 marks)

- All measurement statistics are computed from $p(m) = \text{Tr}(\rho M_m)$, which depends only on ρ . Since $\rho_1 = \rho_2$, no experiment whatsoever can tell the two preparations apart.

- The decomposition of a mixed state into an ensemble is therefore not unique; the density matrix, not the ensemble, is the complete physical description.
- This is why a maximally mixed state carries zero information and why an eavesdropper who holds only half of a Bell pair learns nothing before the classical channel is used.

Question 2.3.3 – NUMERICAL (4 marks)
Quantum Gate Algebra

- Write the matrix of CNOT in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. (1 mark)
- Verify that CNOT is unitary and self-inverse. (1.5 marks)
- Compute the output of CNOT acting on $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ and comment on entanglement. (1.5 marks)

(a) Matrix form (1 mark)

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The upper block is the identity (control = 0); the lower block is X on the target (control = 1).

(b) Unitarity (1.5 marks)

The matrix is real and symmetric, so $\text{CNOT}^\dagger = \text{CNOT}$. Multiplying it by itself, each row picks out exactly one column and the permutation undoes itself:

$$\text{CNOT}^2 = I_4 \quad \Rightarrow \quad \text{CNOT}^\dagger \text{CNOT} = I_4.$$

Hence the gate is unitary and its own inverse. Physically, $|a, a \oplus b\rangle \rightarrow |a, a \oplus a \oplus b\rangle = |a, b\rangle$.

(c) Action on a superposition (1.5 marks)

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$$

The input is separable, $|+\rangle \otimes |0\rangle$, whereas the output cannot be factorized: if $|\Phi^+\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$, then $ad = bc = 0$ while $ac = bd = 1/\sqrt{2}$, a contradiction. The reduced state is $\rho_A = I/2$ with $S_A = 1$ bit, so the gate has generated one ebit of entanglement.

Question 2.3.4 – THEORY (4 marks)

Explain the EPR paradox and how it relates to the completeness of quantum mechanics. Include the EPR criterion of reality, the role of the singlet state, and how Bell's theorem settled the debate experimentally.

Solution:**1. EPR criterion of reality (1 mark)**

EPR (1935) proposed: if, without disturbing a system, the value of a physical quantity can be predicted with certainty, then there exists an element of physical reality corresponding to it. They also assumed locality: a measurement on A cannot instantaneously influence a spacelike-separated B.

2. The singlet argument (1.5 marks)

Take the spin singlet

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

Measuring σ_z on A predicts B's σ_z with certainty; measuring σ_x on A predicts B's σ_x with certainty. By the criterion above, both σ_z and σ_x of B are elements of reality. But quantum mechanics assigns no simultaneous value to non-commuting observables, since $[\sigma_x, \sigma_z] \neq 0$. EPR concluded the wavefunction is an incomplete description, and that hidden variables should supplement it.

3. Bell's theorem (1 mark)

Bell (1964) showed the EPR assumptions are testable. Any local hidden-variable theory obeys the CHSH bound

$$|S| \leq 2, \quad S = E(a, b) - E(a, b') + E(a', b) + E(a', b'),$$

whereas quantum mechanics predicts up to the Tsirelson bound $|S| = 2\sqrt{2} \approx 2.828$ for the singlet with suitable settings.

4. Resolution (0.5 mark)

Experiments from Aspect onwards, and the loophole-free tests of 2015, violate the CHSH inequality in agreement with quantum mechanics. Local realism is therefore untenable. Quantum mechanics is not incomplete; the correlations are non-local, though no-signalling still holds because each local marginal remains $I/2$.

Question 2.3.5 – SCENARIO NUMERICAL (4 marks)**Bell Circuit on IBM Quantum Hardware**

A Bell circuit (H then CNOT) is run on real hardware with 8192 shots. The observed counts are: '00': 3850, '11': 3760, '01': 300, '10': 282.

- (a) State the ideal output distribution. (1 mark)
- (b) Compute the total error rate from the data. (1.5 marks)
- (c) Model the output as $\rho = (1-p)|\Phi^+\rangle\langle\Phi^+| + p I_4/4$ and estimate p and the fidelity. (1.5 marks)

(a) Ideal distribution (1 mark)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \Rightarrow p_{00} = p_{11} = 0.5, \quad p_{01} = p_{10} = 0$$

Expected counts: 4096 each for '00' and '11', zero for the others.

(b) Error rate (1.5 marks)

Counts in states that should never appear:

$$N_{\text{err}} = 300 + 282 = 582$$

$$\varepsilon = \frac{582}{8192} = 0.0710 \quad (7.10\%)$$

Total check: $3850 + 3760 + 582 = 8192$ ✓

(c) Depolarizing estimate (1.5 marks)

Under the depolarizing model the two forbidden outcomes each carry weight $p/4$, so

$$p_{01} + p_{10} = \frac{p}{2} = 0.0710 \quad \Rightarrow \quad p \approx 0.142.$$

Fidelity with the ideal Bell state:

$$F = \langle \Phi^+ | \rho | \Phi^+ \rangle = (1 - p) + \frac{p}{4} = 1 - \frac{3p}{4} = 1 - 0.1065 \approx 0.894.$$

So the prepared state has roughly 89% fidelity. The slight asymmetry between the '00' and '11' counts (3850 vs 3760) points to additional readout bias beyond the symmetric depolarizing channel, which would be corrected by measurement-error mitigation.